

Reviewer Pack — Minimal Order of Twisted Brauer Groups and Communication Com...

Entry 7d8d93876531 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Order of Twisted Brauer Groups and Communication Complexity Rank

Entry ID: 7d8d93876531 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-05 11:13:53 UTC

1. Conjecture

Field A (mathematical branch): Algebra (Twisted Brauer Groups) **Field B** (complexity object): Communication Complexity (Matrix Rank)

Statement:

The communication complexity rank of a circuit is bounded by the minimal order of its corresponding twisted Brauer group, i.e., for any n -input Boolean circuit C , the communication complexity rank $r(C)$ satisfies $r(C) \leq \Theta(\log_2(\text{Oder}(\text{Brauer}(C))))$

Rationale (proposer's reasoning):

Twisted Brauer groups are algebraic structures that can capture non-commutative aspects of representation theory. Their minimal order may provide a direct measure of the complexity inherent in a circuit's structure, which is reflected in its communication complexity rank.

Taxonomy category: communication_complexity (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 6f0c9f6907151703

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all n -input Boolean circuits C , the ratio between the communication complexity rank $r(C)$ and the minimal order of the twisted Brauer group $\log_2(\text{Oder}(\text{Brauer}(C)))$ is less than or equal to 1.5 with 95% confidence. The conjecture is falsified if this ratio exceeds 1.5 for any circuit C .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	1.00	SAFE	SAFE

4. Novelty audit

Verdict: ADJACENT_OK against 3 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "twisted Brauer groups" AND "communication complexity rank" - "minimal order of twisted Brauer groups" AND matrix rank - "Boolean circuit" AND communication complexity AND twisted Brauer group

Top relevant hits considered: - [http://arxiv.org/abs/2503.14117v1] Boolean Circuit Complexity and Two-Dimensional Cover Problems - [http://arxiv.org/abs/2412.14933v1] Cirbo: A New Tool for Boolean Circuit Analysis and Synthesis - [http://arxiv.org/abs/1910.11923v2] Learning Boolean Circuits with Neural Networks

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=124, elapsed=240.4s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_random_circuit(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def communication_complexity(circuit):
        n = len(circuit)
        if n == 1:
            return 1
        return 2 * communication_complexity(circuit[:n//2]) + 1

    def twisted_brauer_group_order(n):
        # Placeholder for actual computation of the twisted Brauer group order
        # This is a dummy implementation that returns a random value for demonstration purposes
        return random.randint(1, 2**n)

    n_values = [5, 10, 15, 20, 30, 40]
    metric_value = 0.0
    instances_tested = 0
    n_max = 0

    for n in n_values:
        if n > n_max:
            n_max = n
        circuit = generate_random_circuit(n)
```

```

        instances_tested += 1
        cc_rank = communication_complexity(circuit)
        order = twisted_brauer_group_order(n)
        metric_value += cc_rank / math.log2(order)

    mean_metric = metric_value / len(n_values)
    conjecture_holds = all(cc_rank <= 1.5 * math.log2(order) for n in n_values for circuit in [gen_circuit(n)])
    counterexample = "" if conjecture_holds else "mapping_undefined"

    return {
        "metric_name": "communication_complexity_ratio",
        "metric_value": mean_metric,
        "instances_tested": instances_tested,
        "n_max": n_max,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [random.randint(2, 1000) for _ in range(30)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_metric = sum(r["metric_value"] for r in results) / len(results)
    std_metric = math.sqrt(sum((r["metric_value"] - mean_metric)**2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric} std={std_metric} support_fraction={support_fraction}")
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"mapping_undefined\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE mapping_undefined")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means we cannot verify the conjecture’s support or falsification. | next: None

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	12484
2	preregistration	ollama_remote	glm4:latest	0	0	9873
3	novelty	ollama_remote	glm4:latest	0	0	8525
4	novelty	ollama_remote	glm4:latest	0	0	9690
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	40980
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10275
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11069
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8989
9	judge	ollama_remote	glm4:latest	0	0	53699

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 165584 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in research/audit/7d8d93876531.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/7d8d93876531.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/7d8d93876531.tar.gz (if generated)